Shaktiranjan Sahoo

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PROFESSIONAL SUMMARY

Azure Data Engineer with 3.5+ years of experience building scalable data pipelines processing multi-source data (SQL Server, ADLS, Blob). Specialized in ADF, Databricks, and PySpark, with proven experience in implementing Medallion architecture and optimizing large-scale ETL pipelines for performance and cost efficiency.

Skills

- Languages: Python, SQL, PySpark
- Big Data: Apache Spark, Delta Lake
- Cloud: Azure (ADF, ADLS Gen2, Databricks, Synapse)
- Data Engineering: ETL/ELT, Data Modeling, SCD, Data Warehousing
- Tools: Azure DevOps, Git,
- Databases: SQL Server, Snowflake
- Monitoring: Azure Monitor, Log Analytics
- Security: Azure Key Vault
- Integration: REST API, HTTP connectors

EXPERIENCE

ALTEN INDIA

Client: ADE(DRDO), Bangalore

Azure Data Engineer Sept 2022 – Present

Built scalable ETL pipelines using Azure Data Factory and Databricks to process multi-source data (~20+ GB/day) into ADLS Gen2, implementing incremental loading (watermark logic) and optimizing storage using Parquet for efficient analytics.

Responsibilities: -

- Designed and implemented scalable ETL pipelines using Azure Data Factory (ADF) to ingest, transform, and load data from multiple sources into Azure Data Lake, Azure Blob Storage, and Azure SQL Database, improving data processing efficiency.
- Developed and managed end-to-end data integration pipelines in ADF using Linked Services, Datasets, and Pipelines, enabling seamless data extraction and consolidation from sources like Azure SQL, Blob Storage, and ADLS Gen2.
- Automated workflow scheduling using triggers, ensuring reliable and efficient data processing across pipelines.
- Built and optimized high-performance Spark applications in Azure Databricks using PySpark and Spark SQL to process and analyze large-scale datasets.
- Implemented Medallion Architecture (Bronze, Silver, Gold layers) to structure data pipelines, enabling incremental data processing and improved data quality for analytics.
- Designed and fine-tuned complex SQL queries to deliver scalable and optimized data solutions, ensuring high performance and alignment with business requirements.
- Collaborated with cross-functional stakeholders to gather requirements, analyze complex data challenges, and deliver data-driven solutions aligned with organizational goals.
- Worked closely with reporting and analytics teams to deliver high-quality, analytics-

ready datasets, enabling better business insights and decision-making.
- Engineered robust CI/CD pipelines using Azure DevOps to automate deployments, manage version control, and ensure seamless integration of data solutions.

Azure Data Migration & Ingestion Framework: -

Built a scalable data migration solution to move data from SQL Server to Azure Data Lake Storage Gen2 using Azure Data Factory. The solution supports incremental data loading using watermark logic and stores data in optimized Parquet format for efficient processing and analytics.

Responsibilities: -
- Designed and implemented end-to-end data migration pipelines using Azure Data Factory to move data from SQL Server to Azure Data Lake Storage Gen2.
- Implemented incremental data loading using watermark (timestamp-based) logic to process only new and updated records, improving efficiency.
- Designed a modular pipeline structure using parent-child pipelines, making the solution reusable and easier to maintain.
- Integrated stored procedures within ADF pipelines to handle dynamic data extraction and incremental load logic.
- Set up Linked Services, Datasets, and Integration Runtime to enable secure and smooth data movement between systems.
- Converted and stored data in Parquet format in ADLS Gen2 to improve performance and optimize storage usage.
- Used parameterization and dynamic content to build flexible pipelines capable of handling multiple tables.
- Added error handling, logging, and retry mechanisms to ensure pipelines run reliably.
- Performed data validation between SQL Server and ADLS Gen2 to ensure accuracy and consistency.
- Automated pipeline execution using ADF triggers and scheduling.

Automated DO-178B Requirements Traceability Tool (Python + GUI): -

- Developed an internal automation tool to support DO-178B traceability and verification processes.
- Built a Python-based GUI application to automate traceability across SRD, FRD, CSU/HSI test procedures, test reports, inspection documents, and source code, ensuring compliance with DO-178B Level A objectives.
- Implemented logic to extract Requirement Identifiers (RI), verify RI presence a missing or incomplete traceability links, supporting bi-directional traceability (Req ↔ Test, Req ↔ Code).
- Automated generation of a structured Excel Traceability Report, including:
- Cover Sheet: summary of total requirements, traced/untraced RIs, missing RI documents, software version, and timestamp.
- Traceability Matrix: system requirement ID, RI mapping, verification method (CSU/HSI), verification level, linked test procedures/reports, author details, modification dates, and overall pass/fail status.
- Enhanced certification readiness by improving accuracy of requirement-to-test linkage and significantly reducing manual traceability preparation effort for DO-178B audits and reviews.

CERTIFICATIONS

- Microsoft Fabric Analytics Engineer Associate (DP-700) in progress.
- Microsoft Certified: Azure Data Fundamentals (DP- 900) Microsoft Certified: Azure Data Engineer Associate (DP-203) in progress.

KEY ACHIEVEMENTS

- Reduced ETL pipeline runtime by 35% using Spark optimization techniques
- Designed scalable data pipeline handling 50+ tables with dynamic parameterization
- Improved data quality using Medallion architecture (Bronze-Silver-Gold)

EDUCATION

Biju Patnaik University of Technology (BPUT)
Odisha *2018-2022*

- Bachelor of Engineering in Computer Science (CGPA: 8.77)